Canonical Bundle Formula

We work over an algebraically closed field of characteristic zero, like $\mathbb{C}$. All varieties are assumed to be normal unless otherwise stated.

1 Birational divisors

Definition 1 (b-divisors)**.** Let X be a normal algebraic variety. We call Y a *birational model over* X, or simply a *model over* X, if there exists a proper birational morphism $Y \to X$ of normal varieties. When both X and Y are quasi-projective, $Y \to X$ is projective.

Let $X' \dashrightarrow X$ be a birational map. For any prime divisor E (resp. algebraic valuation v) over X, we denote by $c_{X'}(E)$ (resp. $v_{X'}$) the centre of E (resp. v) on X', whenever it exists. A *b-divisor* $\mathbf{D}$ on X is a formal sum

$$\mathbf{D} = \sum_E r_E E \qquad \left(\text{resp. } \sum_v r_v v\right),$$

where the E are prime divisors over X (resp. the v are algebraic valuations over X) and $r_E \in \mathbb{R}$ (resp. $r_v \in \mathbb{R}$), such that only finitely many terms with nonzero coefficient have a divisorial centre on X. If, in addition, $r_E \in \mathbb{Q}$ for every E (resp. $r_v \in \mathbb{Q}$ for every v), then $\mathbf{D}$ is called a $\mathbb{Q}$-*b-divisor* on X.

The *trace* of $\mathbf{D}$ on a model $X' \to X$ is the $\mathbb{R}$-divisor

$$\operatorname{tr}_{X'} \mathbf{D} = D_{X'} = \sum_{E' \text{ a divisor on } X'} r_{E'} E' \qquad \left(\text{resp. } \sum_{v_{X'} \text{ a divisor}} r_v v_{X'}\right).$$

If $D_{X'}$ is $\mathbb{R}$-Cartier and D_Y is the pullback of $D_{X'}$ on Y for every model $Y \to X'$, we say that $\mathbf{D}$ *descends to* X' and write

$$\mathbf{D} = \overline{D_{X'}}.$$

Let $X \to U$ be a proper morphism, and assume that $\mathbf{D}$ is a b-divisor over X which descends to some model $Y \to X$. If D_Y is nef$/U$ (resp. semi-ample$/U$), then we say that $\mathbf{D}$ is *b-nef*$/U$ (resp. *b-semi-ample*$/U$). If D_Y is an $\mathbb{R}$-Cartier divisor, then we say that $\mathbf{D}$ is $\mathbb{R}$-*b-Cartier*. The terms b-Cartier and $\mathbb{Q}$-b-Cartier are defined similarly using Cartier and $\mathbb{Q}$-Cartier divisors, respectively. If $\mathbf{D}$ can be written as an $\mathbb{R}_{\geq 0}$-linear combination of nef$/U$ $\mathbb{Q}$-b-Cartier b-divisors, then we say that $\mathbf{D}$ is NQC/U.

When $U = \operatorname{Spec} k$, this gives the absolute setting.

Definition 2 (Generalized pairs)**.** A *generalized sub-pair* (or *g-sub-pair* for short) is a triple $(X, B, \mathbf{M})/U$ consisting of a normal algebraic variety X associated with a proper morphism $X \to U$, an $\mathbb{R}$-divisor B on X, and an $\mathbb{R}$-b-Cartier b-nef$/U$ divisor $\mathbf{M}$ on X, such that $K_X + B + M_X$ is $\mathbb{R}$-Cartier. Note that, on a model $h : W \to X$ on which $\mathbf{M}$ descends, we put

$$M_X := h_* M_W.$$

If B is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is $\mathbb{Q}$-b-Cartier, then we say that $(X, B, \mathbf{M})/U$ is a $\mathbb{Q}$-*g-sub-pair*.

If $\mathbf{M} = 0 = \overline{0}$, a g-sub-pair $(X, B, \mathbf{M})/U$ is called a *sub-pair*. If $U = \{\text{pt}\}$, or if U is not important, we drop U from the notation.

A g-sub-pair (resp. $\mathbb{Q}$-g-sub-pair) $(X, B, \mathbf{M})/U$ is called a *g-pair* (resp. $\mathbb{Q}$-*g-pair*) if $B \geq 0$. A sub-pair (X, B) is called a *pair* if $B \geq 0$.

Definition 3 (Singularities of g-pairs)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For any prime divisor E on X and any $\mathbb{R}$-divisor D on X, we define $\operatorname{mult}_E D$ to be the multiplicity of D along E.

Let $h : W \to X$ be any log resolution of (X, B) such that $\mathbf{M}$ descends to W, and write

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

The *log discrepancy* of a prime divisor E on W with respect to $(X, B, \mathbf{M})$ is $1 - \operatorname{mult}_E B_W$, and it is denoted by $A(E, X, B, \mathbf{M})$. For the discrepancy

$$a(E, X, B, \mathbf{M}) = -\operatorname{mult}_E B_W,$$

we have

$$A(E, X, B, \mathbf{M}) = 1 + a(E, X, B, \mathbf{M}).$$

We say that $(X, B, \mathbf{M})$ is *sub-g-lc* (resp. *sub-g-klt*) if

$$A(E, X, B, \mathbf{M}) \geq 0 \qquad (\text{resp. } > 0)$$

for every log resolution $h : W \to X$ as above and every prime divisor E on W.

Suppose that $(X, B, \mathbf{M})$ is sub-g-lc. A *g-lc place* of $(X, B, \mathbf{M})$ is a prime divisor E over X such that

$$A(E, X, B, \mathbf{M}) = 0.$$

A *g-lc centre* of $(X, B, \mathbf{M})$ is the centre of a g-lc place on X. We often say *lc place* and *lc centre* for short when the pair is clear.

We say that $(X, B, \mathbf{M})$ is *(sub-)g-dlt* if it is (sub-)g-lc and there exists an open subset $U \subseteq X$ such that

(1) $(U, B|_U)$ is log smooth;

(2) for any lc place E over X, we have $c_X(E) \cap U \neq \varnothing$, and $c_X(E) \cap U$ is an lc centre of $(U, B|_U)$.

There is an important result in [Bir-Zhang14, proof of Lemma 4.5], also proved in [HL22, Proposition 3.9].

Definition–Lemma 4 (G-dlt modifications)**.** Let $(X, B, \mathbf{M})/U$ be a quasi-projective g-lc g-pair. Then there is a projective birational morphism $r : X' \to X$ and a g-lc g-pair $(X', B', \mathbf{M})/U$ such that

(1) $(X', B', \mathbf{M})$ is $\mathbb{Q}$-factorial g-dlt;

(2) $K_{X'} + B' + M_{X'} = r^*(K_X + B + M_X)$; and

(3) any r-exceptional divisor is a component of $\lfloor B' \rfloor$.

For any projective birational morphism r and $(X', B', \mathbf{M})/U$ which satisfy (1)–(3), $(r, (X', B', \mathbf{M})/U)$ will be called a *g-dlt modification* of $(X, B, \mathbf{M})$.

It has a useful application; see [JLX22, Lemma 2.6] or my note [Adj, 9, 10, 14, 15].

Definition–Lemma 5 (G-dlt adjunction)**.** Let $(X, B, \mathbf{M})/U$ be a g-lc g-pair and let $S \subseteq \lfloor B \rfloor$ be a prime divisor, with normalization $\nu : S^\nu \to S$. Then there is a naturally defined g-pair structure $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ such that

$$K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu} \sim_{\mathbb{R}} (K_X + B + M_X)|_{S^\nu}.$$

We say that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ is the g-pair obtained by *divisorial adjunction*. If $(X, B, \mathbf{M})$ is g-lc, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-lc, and if $(X, B, \mathbf{M})$ is g-dlt, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-dlt.

For the construction of divisorial adjunction, see [Adj, 9 and 10]; for the g-dlt and g-lc parts, see [Adj, 14 and 15].

Definition 6 (Discrepancy b-divisors)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For each model $f : Y \to X$, choose compatible canonical divisors with $f_*K_Y = K_X$, and set

$$\operatorname{tr}_Y \mathbf{A}(X, B, \mathbf{M}) = K_Y + M_Y - f^*(K_X + B + M_X).$$

These traces define the *discrepancy b-divisor* $\mathbf{A}(X, B, \mathbf{M})$; we often write its trace A_Y for short. Define another b-divisor by

$$\operatorname{tr}_Y \mathbf{A}^*(X, B, \mathbf{M}) = A_Y^{>-1},$$

where we omit the components with coefficient at most -1. For a prime divisor E over X,

$$\operatorname{mult}_E \mathbf{A}(X, B, \mathbf{M}) = A(E, X, B, \mathbf{M}) - 1 = a(E, X, B, \mathbf{M}),$$

where $a(E, X, B, \mathbf{M})$ is the discrepancy of E.

Definition 7 (Sheaf associated to a b-divisor)**.** Let $\mathbf{D}$ be an $\mathbb{R}$-b-divisor on X. The sheaf $\mathcal{O}_X(\mathbf{D})$ is an $\mathcal{O}_X$-module whose sections on an open subset $U \subseteq X$ are given by

$$H^0\big(U, \mathcal{O}_X(\mathbf{D})\big) = \big\{\varphi \in k(X)^* \;\big|\; \operatorname{div}_Y(\varphi) + D_Y \geq 0 \text{ on } h^{-1}(U) \text{ for every model } h : Y \to X\big\} \cup \{0\},$$

where $k(X)$ is the function field of X. Note that $\mathcal{O}_X(\mathbf{D})$ is not necessarily coherent.

Lemma 8. *When* $\mathbf{D}$ *is b-Cartier,* $\mathcal{O}_X(\mathbf{D})$ *is coherent.*

Proof. Take a model $h : W \to X$ on which $\mathbf{D}$ descends. Then we have

$$\mathcal{O}_X(\mathbf{D}) = h_*\mathcal{O}_W(D_W).$$

Indeed, the inclusion $\mathcal{O}_X(\mathbf{D}) \subseteq h_*\mathcal{O}_W(D_W)$ is trivial. For the other inclusion, suppose that $\varphi \in H^0(h^{-1}(V), \mathcal{O}_W(D_W))$. This means that

$$\operatorname{div}_W(\varphi) + D_W \geq 0 \qquad \text{on } h^{-1}(V).$$

For any other model $Y \to X$, take a high common model

$$W \xleftarrow{\ r\ } T \xrightarrow{\ s\ } Y$$

over X. Since $\mathbf{D}$ descends to W,

$$\operatorname{div}_T(\varphi) + D_T = r^*\big(\operatorname{div}_W(\varphi) + D_W\big) \geq 0$$

on $(h \circ r)^{-1}(V)$. Pushing forward via s gives

$$\operatorname{div}_Y(\varphi) + D_Y = s_*\big(\operatorname{div}_T(\varphi) + D_T\big) \geq 0$$

on $s((h \circ r)^{-1}(V))$. Thus $h_*\mathcal{O}_W(D_W) \subseteq \mathcal{O}_X(\mathbf{D})$, and coherence follows from the properness of h. □

For a b-divisor $\mathbf{D}$ on X, its *round up* $\lceil \mathbf{D} \rceil$ is defined by

$$\mathrm{tr}_Y \lceil \mathbf{D} \rceil = \lceil D_Y \rceil$$

on every model $Y \to X$. Similarly for round down.

Definition 9 (Contraction)**.** A *contraction* is a proper surjective morphism $f : X \to Y$ of normal varieties with $f_* \mathcal{O}_X = \mathcal{O}_Y$.

Equivalently, f has connected fibres. Indeed, in the Stein factorization

$$X \longrightarrow \widetilde{Y} \longrightarrow Y,$$

connected fibres force the finite morphism $\widetilde{Y} \to Y$ to have degree one, i.e., it is a birational morphism between normal varieties, so it is an isomorphism.

Definition 10 (G-lc-trivial fibration)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over U. We call $f : (X, B, \mathbf{M}) \to Y$ a *g-lc-trivial fibration* over U if

(1) $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y;

(2) $\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = 1$; and

(3) there exists an $\mathbb{R}$-Cartier divisor L on Y such that

$$K_X + B + M_X \sim_{\mathbb{R}} f^* L.$$

If, in addition, $(X, B, \mathbf{M})$ is g-sub-klt over the generic point of Y, we say *g-klt-trivial*. For $\mathbf{M} = 0$, these are classical lc-trivial and klt-trivial fibrations; see [FG14, Definitions 3.1–3.2].

Remark 11. Condition (3) is sometimes written as

$$K_X + B + M_X \sim_{\mathbb{R}, Y} 0.$$

When f is a contraction, they are equivalent.

Lemma 12 ([Fuj12, Lemmas 3.19–3.20])**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-lc g-sub-pair, and let $h : W \to X$ be a log resolution on which $\mathbf{M}$ descends. Write*

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

Then

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

In particular, this sheaf is coherent. If $B \geq 0$, then

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \mathcal{O}_X.$$

Proof. By definition,

$$\mathrm{tr}_W \, \mathbf{A}(X, B, \mathbf{M}) = -B_W, \qquad \mathrm{tr}_W \, \mathbf{A}^*(X, B, \mathbf{M}) = -B_W^{<1}.$$

Let $g : W' \to W$ be any model over W. Define $B_{W'}$ by

$$K_{W'} + B_{W'} + M_{W'} = (h \circ g)^*(K_X + B + M_X).$$

Since $\mathbf{M}$ descends to W, we have $M_{W'} = g^* M_W$. Cancelling these terms gives

$$K_{W'} + B_{W'} = g^*(K_W + B_W).$$

Step 1. We claim that

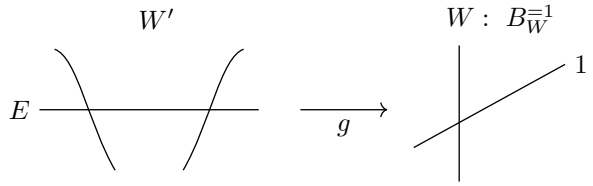
$$\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil = g^* \lceil \mathbf{A}^*(X, B, \mathbf{M})_W \rceil + F,$$

where $F \geq 0$ is an integral g-exceptional divisor.

The divisor $\lceil -B_W^{<1} \rceil$ is an effective Cartier divisor, and is supported on the negative components of B_W. Thus the pair

$$\big(W, B_W + \lceil -B_W^{<1} \rceil\big)$$

is log smooth and has lc singularities, since for a coefficient $b < 1$, we have $0 \leq b + \lceil -b \rceil < 1$. It also has the same lc places as (W, B_W), because an lc place of either log smooth pair has centre a stratum of $B_W^{=1}$, but $\operatorname{Supp} \lceil -B_W^{<1} \rceil$ contains no such stratum.



$$a(E) = 2 - 1 - (1 + 1) = -1$$

For a prime divisor E on W', the number $\operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil$ is a nonnegative integer, since $g^* \lceil -B_W^{<1} \rceil \geq 0$. If $\operatorname{mult}_E B_{W'} = 1$, then E is an lc place of (W, B_W), so its multiplicity in $g^* \lceil -B_W^{<1} \rceil$ is zero. Both sides of the claimed equality

$$\lceil -B_{W'}^{<1} \rceil = g^* \lceil -B_W^{<1} \rceil + F$$

then give $\operatorname{mult}_E F = 0$.

If $\operatorname{mult}_E B_{W'} < 1$, we have

$$A(E, W, B_W) = 1 - \operatorname{mult}_E B_{W'} > 0,$$

so E is not an lc place of (W, B_W). Since the two pairs have the same lc places, we have

$$\begin{aligned} 0 &< A\big(E, W, B_W + \lceil -B_W^{<1} \rceil\big) \\ &= 1 - \operatorname{mult}_E B_{W'} - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil. \end{aligned}$$

Rounding this inequality gives

$$\lceil -\operatorname{mult}_E B_{W'} \rceil \geq \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil \geq 0.$$

The left-hand side is nothing but $\operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil$, so $\operatorname{mult}_E F \geq 0$. Since (W, B_W) is sub-lc, the coefficients of $B_{W'}$ are at most 1. Therefore $F \geq 0$ is effective. Since both sides of the claimed equality are integral divisors, F is integral.

Finally, if E is not g-exceptional, then $g(E)$ is a divisor on W. That is, g is an isomorphism over the generic point of $g(E)$ by normality, so we have

$$\operatorname{mult}_E B_{W'} = \operatorname{mult}_{g(E)} B_W.$$

Then

$$\operatorname{mult}_E F = \operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil = 0.$$

We see that F is g-exceptional. This proves the claim.

Step 2. By algebraic Hartogs, we have

$$g_* \mathcal{O}_{W'}(F) = \mathcal{O}_W.$$

Then we have

$$\begin{aligned} g_* \mathcal{O}_{W'}\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil\big) &= g_* \mathcal{O}_{W'}\big(g^* \lceil -B_W^{<1} \rceil + F\big) \\ &\cong \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big), \end{aligned}$$

thanks to the projection formula. These identifications are induced by the identity on rational functions.

For any other model $\psi : Y \to X$, take a common higher model

$$W \xleftarrow{r} T \xrightarrow{s} Y.$$

Fix any open subset $V \subseteq X$. Suppose that we have

$$\operatorname{div}_W(\varphi) + \lceil -B_W^{<1} \rceil \geq 0$$

on $h^{-1}(V)$. Applying the claimed equality to $r : T \to W$, we have

$$r_* \mathcal{O}_T\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \cong \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

Thus it gives

$$\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil \geq 0$$

on $(h \circ r)^{-1}(V)$. We push forward this along $s : T \to Y$ and get

$$\operatorname{div}_Y(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_Y \rceil = s_*\big(\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \geq 0$$

on $s((h \circ r)^{-1}(V)) = \psi^{-1}(V)$. It gives

$$h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big) \subseteq \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big).$$

The reverse direction is trivial. Coherence is due to the properness of h.

Finally, if $B \geq 0$, then every non-exceptional component of B_W has coefficient in $[0, 1]$. Hence $\lceil -B_W^{<1} \rceil$ is effective and h-exceptional. By algebraic Hartogs, we have

$$h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big) = \mathcal{O}_X.$$

□

Now let $f : X \to Y$ be a contraction over U. If $(X, B, \mathbf{M})$ is only g-sub-lc over the generic point of Y, i.e., it is g-sub-lc over an open subset $Y^0 \subseteq Y$, applying Lemma 12, we have

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big)$$

on $f^{-1}(Y^0)$. Pushing forward along f, we get

$$f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = (f \circ h)_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big)$$

on Y^0, and since $\eta_Y \in Y^0$, we have

$$\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \operatorname{rank}(f \circ h)_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

Lemma 13 (The rank-one condition)**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over U. Suppose that $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y and that B is effective over that point, i.e., its components which dominate Y have nonnegative coefficients. Then*

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = 1.$$

Proof. We may replace Y by an open subset and assume that $B \geq 0$ and $(X, B, \mathbf{M})$ is g-lc; this does not change the rank. Take $h : W \to X$ to be a log resolution on which $\mathbf{M}$ descends. The divisor $\lceil -B_W^{<1}\rceil$ is effective and h-exceptional. So by Lemma 12,

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = \operatorname{rank} f_*\mathcal{O}_X = \operatorname{rank} \mathcal{O}_Y = 1.$$

□

Example 14 ($B \not\geq 0$ over the generic point)**.** Put $f : \mathbb{P}^1 \to \operatorname{Spec} \mathbb{C}$, $B = -p$, and $\mathbf{M} = 0$, where $p \in \mathbb{P}^1$. The pair $(\mathbb{P}^1, B)$ is a sub-klt pair. Take $h = \mathrm{id}$. Then

$$h_*\mathcal{O}_{\mathbb{P}^1}\big(\lceil -B^{<1}\rceil\big) = \mathcal{O}_{\mathbb{P}^1}(p).$$

Therefore,

$$\operatorname{rank} f_*\mathcal{O}_{\mathbb{P}^1}(p) = \dim_{\mathbb{C}} H^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(p)\big) = 2,$$

not rank one.

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